

Yangruirui ZHOU

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EDUCATION

Boston University, Boston, MA

09/2020-01/2026

Doctor of Philosophy (Ph.D.) in Computer Engineering

GPA: 3.91/4.00

Advisor: Douglas Densmore

Thesis: AI-driven design automation for microfluidics and multicellular systems

Research Focus: AI-enabled microfluidic CAD, formal verification, and multicellular biological system design.

University of Electronic Science and Technology of China (UESTC), Chengdu, China

09/2016-06/2020

Bachelor of Engineering in Software Engineering (*Elite program*)

GPA: 3.94/4.00 (Rank: 6/740)

Thesis: Analysis of Postoperative Applications of Pose Tracking Algorithms.

RESEARCH INTERESTS

- [1] AI-driven closed-loop Design-Build-Test-Learn workflow for microfluidic systems
- [2] Constraint-based optimization and graph algorithms for multicellular computation
- [3] Formal verification and CAD software infrastructures for biological design
- [4] Large language model (LLM)-enabled next generation biological design automation
- [5] Self-driving laboratories for programmable biological systems

AWARDS & HONORS

iGEM "Gold medal" (Student Mentor, wiki: https://2024.igem.wiki/bostonu/)	11/2024
China Youth Science and Technology Innovation Award (Nominated by UESTC)	05/2020
Most Outstanding Students of UESTC 2019 (成电杰出学生, 10/5000 in UESTC, 1/740 in department)	12/2019
iGEM "Gold medal" and "Best Software Project" (wiki: https://2019.igem.org/Team:UESTC-Software)	11/2019
Outstanding Graduates Award of UESTC	10/2019
"Wu Liang Ye" Enterprise Scholarship (2/740 in Software Engineering department)	09/2019

TECHNICAL SKILLS

AI & Data Science: Graph Algorithm Library (NetworkX), ML/AI/DS Library (PyTorch, TensorFlow, NumPy)

Systems & Software: Python, C/C++, Java, TypeScript, JavaScript, HTML, Linux Shell, LaTeX

Biological & Microfluidic Skills: CNC Milling, Laser Cutting, Benchmark Collection and Evaluation

CAD Tools: Fusion 360, Microfluidic Devices Manual Design

RESEARCH EXPERIENCE

Research Assistant | Boston University, CIDAR lab

Boston, MA | 09/2020-Present

Supervisor: Prof. Douglas Densmore

- **Neptune — Hierarchical Microfluidic Design Automation System (CUHK collaboration)**
 - Developed a full-stack microfluidic design automation system enabling end-to-end synthesis from high-level behavioral specifications to physical layout generation.
 - Developed a DSL and compiler that transform behavioral design inputs into structured connectivity graphs, which are further processed by external placement and routing algorithms; the DSL is currently supported by all five major mainstream large language models for assisted program synthesis.

- Built a web-based interface tightly integrated into the lab's existing 3DuF design environment, enabling direct visualization, interactive editing, and manual refinement of synthesized microfluidic designs.
- **Oriole — Genetic Circuit Partitioning for Multicellular Computation (MIT collaboration)**
 - Developed a sub-graph partitioning algorithm enabling implementation of complex computational functions (e.g., cryptographic hashing) in multicellular systems under biological and hardware constraints.
 - Designed a constraint formalization layer that encodes biological and experimental constraints into computational representations, enabling automated design and optimization in genetic circuit synthesis.
- **Vespa — In-silico Logic-level Constraint-based Validation for Large-scale Microfluidic Systems**
 - Developed an automated in-silico validation system for microfluidic workflows, enabling simulation-based verification and reliability testing prior to wet-lab execution.
 - Constructed benchmark datasets and logic-level constraint representations for evaluating and optimizing microfluidic design automation methods.
- **Other Works**
 - Designed and fabricated modular microfluidic systems supporting multi-modal experimental workflows for studying cell–cell communication in controlled environments.
 - Built a natural language interface using large language models (Claude, GPT, Gemini, Qwen, DeepSeek) enabling users to interact with microfluidic design workflows without programming.

iGEM Team Student Mentor | BostonU (Wiki: <https://2024.igem.wiki/bostonu/>) Boston, MA | 05-10/2024

- Led computational and software design for iGEM 2024 project, bridging biological objectives with system-level modeling and implementation.
- Mentored students in system design and problem decomposition for biological engineering workflows.

iGEM Team Member | UESTC (Wiki: <https://2019.igem.org/Team:UESTC-Software>) Chengdu, China | 05-10/2019

- Built a cross-database sequence-to-protein mapping pipeline improving annotation between iGEM Registry and UniProt.
- Developed a ranking and filtering model improving alignment accuracy to >95% and increasing retrieval coverage by ~400%.

Student Research Internship | UCSB, Four “I”s lab Santa Barbara, CA | 06-08/2019

Supervisor: Prof. Matthew Turk and Prof. Tobies Hollerer

- Reimplemented a state-of-the-art pose tracking algorithm for local evaluation and deployment.
- Benchmarked performance against a novel method using standardized datasets.

PROFESSIONAL SERVICES & ACTIVITIES

Mentor, EBRC Mentorship for Undergraduate and Master's Students Program (EMUMS)	04/2025-Now
Direct Mentor, Eric Xie , Undergraduate Research Assistant, CIDAR Lab	06/2023-Now
International Workshop on Bio-Design Automation (IWBD'A25), Oral Presentation	09/2025
International Genetically Engineered Machine competition (iGEM'19, iGEM'24)	11/2019, 11/2024
Inventor/Contributor, Patent: Agricultural Heavy Metal Biosensor-Integrated Device	05-11/2024
Student Mentor of iGEM team: BostonU	04-10/2024
Chicago BioEngineering Conference (CBEC'24)	09/2024
Teaching Assistant, EC504: Advanced Data Structures and Algorithms , Boston University	01-05/2022, 01-05/2023
Engineering Biology Research Consortium (EBRC'23)	06/2023
International Workshop on Bio-Design Automation (IWBD'A22), Poster Presentation	10/2022

PROFESSIONAL SOCIETIES

Institute of Electrical and Electronics Engineers (IEEE, graduate student membership)	02/2024-Now
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JOURNAL REVIEWER

IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (IEEE-TCAD)

eLife

International Conference on Computer-Aided Design (ICCAD'22)

PUBLICATIONS

Google Scholar (<https://scholar.google.com/citations?user=Zel-iSQAAAAJ&hl=en>)

Journals:

[J1, IEEE-TCAD 44.2 (2024): 483-496.] **Zhou, Y.**, Oliveira, S. M., Sanka, R., McIntyre, D., & Densmore, D. (2024). Vespa: Logic-Level Constraint-Based Validation for Continuous-Flow Microfluidic Devices.

[J2, Nature Chemical Biology 21.2 (2025): 268-279.] Padmakumar, J. P., Sun, J. J., Cho, W., **Zhou, Y.**, Krenz, C., Han, W. Z., ... & Voigt, C. A. (2024). Partitioning of a 2-bit hash function across 66 communicating cells.

[J3, IEEE-TCBB 22.4 (2025): 1888-1901.] **Zhou, Y.**, Voigt, C. A., & Densmore, D. (2025). Constraint-Based Sub-Graph Partitioning for Multi-Cellular Biological Networks.

Conferences & Workshops:

[C1, IWBDA'22] A Conceptual Interactive Microfluidic Design and Control Workflow

[C2, IWBDA'25] LaNVis: Biological Constraint-based Large Network Visualizer (Accepted for oral presentation)

Manuscripts in Preparation:

[J4, in progress, targeting Nature Computational Science] *Neptune: An open-source platform for microfluidic design automation with a new programming language and in-context LLM assistance.*